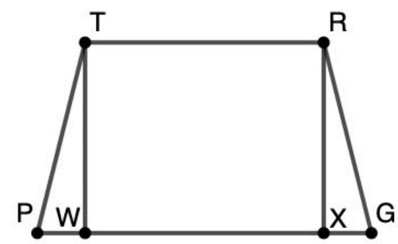
Geometry
Unit 8
Lesson 5 Homework

Name Answer Key__________
Date_____________________
Period___________________

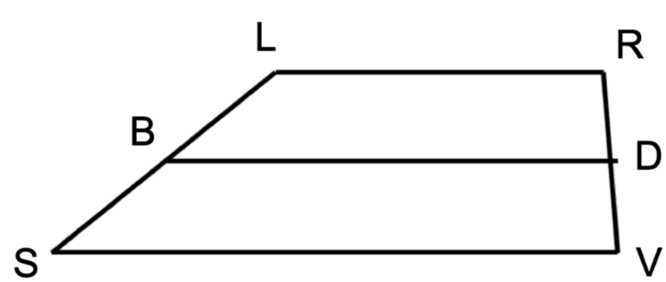


Given: $TRGP$ is an isosceles trapezoid with altitudes $\overline{TW}$ and $\overline{RX}$.

Prove: $\overline{PW} \cong \overline{XG}$

Statement	Reason
$TRGP$ is an isosceles trapezoid	Given
1. $\overline{TW} \cong \overline{RX}$	Altitudes of an isosceles trapezoid are congruent.
$\overline{TW} \perp \overline{PG}$ and $\overline{RX} \perp \overline{PG}$	2. Definition of altitude
$\angle TWP$ and $\angle RXG$ are right angles.	Perpendicular lines form right angles
3. $\angle TWP \cong \angle RXG$	All right angles are congruent
4. $\angle TPW \cong \angle RGX$	5. Base angles of an isosceles trapezoid are congruent
6. $\Delta TPW \cong \Delta RGX$	AAS Congruence Theorem
$\overline{PW} \cong \overline{XG}$	Corresponding parts of congruent triangles are congruent

Trapezoid $LRVS$ is shown with the midsegment $\overline{BD}$ where $LB = 5 + 3x$, $SB = 5x - 11$, $RD = 9y + 16$, $VD = 51 + 4y$.



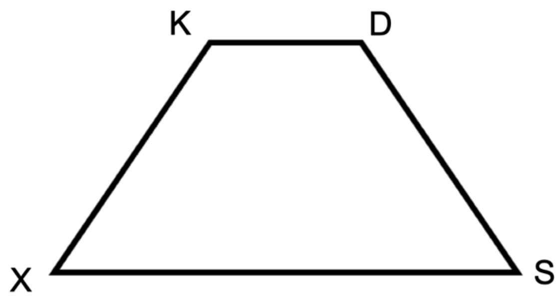
7. What is the value of x and y?

$x = 8$
$y = 7$

8. Find the length of $\overline{LB}$ and $\overline{RV}$.

$LB = 29$
$RV = 158$

$KDSX$ is an isosceles trapezoid, as shown, where $m\angle K = (35z - 20)°$, $m\angle X = (15z + 8)°$, $m\angle S = (10t + 25)°$.



9. What is the value of z and t? Round your answer to the nearest hundredth.

$z = 3.84$
$t = 4.06$

10. What is $m\angle S$ and $m\angle D$? Round your answer to the nearest tenth.

$m\angle S = 65.6°$
$m\angle D = 114.4°$